

# #Atmospheric Dynamics

## *Chapter Two: Fundamental Forces and Their Derivation*



### *Learning outcome*

- To Define the *Pressure Gradient Force and derive its equation*
- To Be acquainted with the *Gravitational Force and its derivation.*

# ***Fundamental Forces***

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- The forces that influence atmospheric motions can be classified as either *body forces* or *surface forces*
- **Body forces** act on the center of mass of a fluid parcel; they have magnitudes proportional to the mass of the parcel. They act at a distance and affect the entire fluid volume; e.g. *Gravity*
- **Surface forces** act across the boundary surface separating a fluid parcel from its surroundings; their magnitudes are independent of the mass of the parcel. They act locally upon a part of a fluid. They also have direct mechanical contact between two bodies.  
e.g. *pressure gradient force*

# *Fundamental Forces*

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- ✓ For atmospheric motions, the forces that are of primary concern are *the pressure gradient force*, *the gravitational force*, and *friction or Viscous force*.
- ✓ For a coordinate system rotating with the earth, apparent forces are introduced: *the centrifugal force* and *the Coriolis force*.

# ***Pressure Gradient Force***

## *Pressure Gradient Force*

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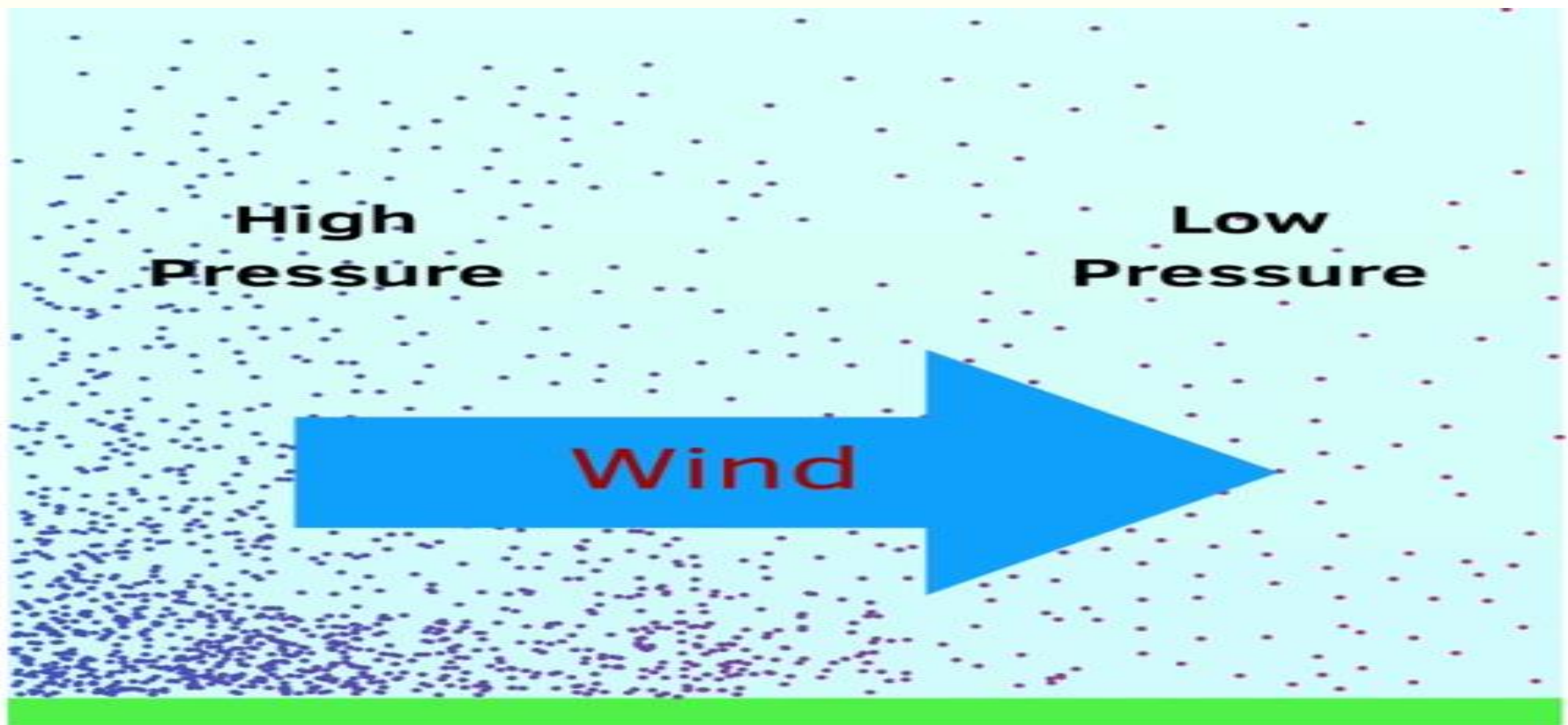
- ❑ Pressure Gradient Force (PGF): the force applied to a small parcel of air *due to pressure differences* created by a difference in air density (more molecules), air temperature (more energy per molecular collision), or both.
- ❑ PGF is always perpendicular to the isobars, with a direction from high pressure toward low pressure.

□ Pressure Gradient: the rate at which pressure changes with distance the most *extreme* horizontal pressure gradients on earth occur in *tornadoes* and in the *eyewalls of hurricanes*, and in rare cases, in extratropical cyclones.

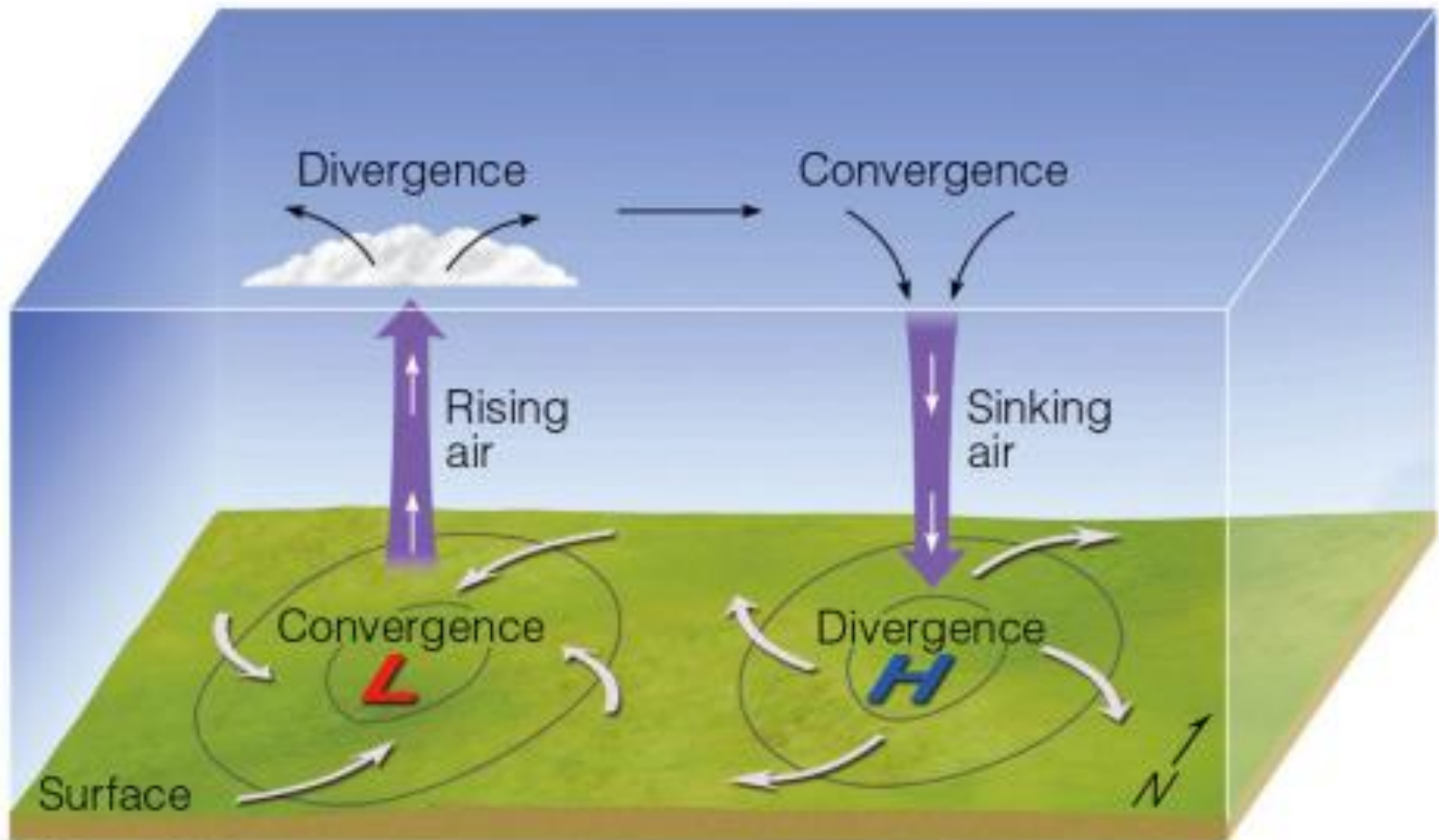
## *Pressure Gradient Force*

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- ❑ A pressure force can only be exerted within a fluid if there is a difference in pressure from one parcel to another; otherwise the parcels are exerting an equal and opposite force on each other, and there is no net force.
- ❑ Therefore, the force associated with pressure arises from the pressure gradient, not the pressure itself.

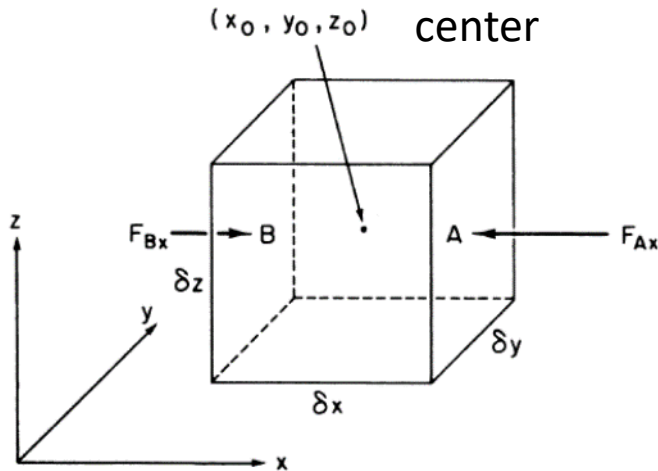






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# Pressure Gradient Force derivation



$$\delta V = \delta x \delta y \delta z \dots \text{Small volume}$$

- The momentum transfer per unit time per unit area is the pressure exerted on the walls of the volume element by the surrounding air.

infinitesimal volume element of air (air parcel)

- Using Taylor series expansion, the pressure on the wall labelled A is:

$$p_0 + \frac{\partial p}{\partial x} \frac{\delta x}{2} + \text{higher order terms}$$

- $p_0$  is pressure at the center

## *Pressure Gradient Force derivation*

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- Neglecting the higher order terms in this expansion, the pressure force acting on the volume element at wall A is:

$$F_{Ax} = - \left( p_0 + \frac{\partial p}{\partial x} \frac{\delta x}{2} \right) \delta y \delta z \text{ (Force = pressure X area)}$$

$\delta y \delta z$  is the area of wall A

- Similarly, the pressure force acting on the volume element at wall B is just

$$F_{Bx} = \left( p_0 - \frac{\partial p}{\partial x} \frac{\delta x}{2} \right) \delta y \delta z$$

- Therefore, the net x component of this force acting on the volume is

$$F_X = F_{Ax} + F_{Bx} = - \frac{\partial p}{\partial x} \delta x \delta y \delta z$$

## ***Pressure Gradient Force derivation***

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- The mass  $m$  of the differential volume element is simply the density  $\rho$  times the volume:  $m = \rho \delta x \delta y \delta z$
- Thus, the x component of the pressure gradient force per unit mass is:

$$\frac{F_x}{m} = -\frac{1}{\rho} \frac{\partial p}{\partial x}$$

- Similarly, it can easily be shown that the y and z components of the pressure gradient force per unit mass are:

$$\frac{F_y}{m} = -\frac{1}{\rho} \frac{\partial p}{\partial y} \quad \text{and} \quad \frac{F_z}{m} = -\frac{1}{\rho} \frac{\partial p}{\partial z}$$

- so that the total pressure gradient force per unit mass is:

$$\frac{\mathbf{F}}{m} = -\frac{1}{\rho} \nabla p$$

# Exercise

□ let the surface pressure at station A be 1001.5hpa, while the surface pressure at station B 100km west of station A is 1000.5hpa assuming air density to be  $1.2\text{kg/m}^3$ , and find the pressure gradient force per unit mass. It is assumed that both stations have the same elevation

# Gravitational Force

- The force due to gravity originates from the ***mutual attraction between any object or fluid parcel and the Earth.***
- Detailed observations of periodic motions in the solar system confirm the law, developed by Newton and published in his *Principia* in 1687, that every particle in the Universe attracts every other particle with a force proportional to the product of the masses and inversely proportional to the square of the distance between them.
- The universal constant of proportionality is
$$G = 6.673 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}.$$

# **How does gravity affect the wind?**

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- ❑ The most basic answer to our question is that gravity is necessary in order to hold our atmosphere to the planet. Air molecules are in constant motion at the molecular level, like a room full of bugs flying in every direction.
- ❑ The typical air molecule moves about 500m/s (1100mph) in air at 27°C (about 80°F), but only for a very short distance before bumping into something or another air molecule.
- ❑ At the outer edge of our atmosphere those molecules moving outward, without anything with which to collide, could escape from the Earth.

# How does gravity affect the wind?

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- ❑ Some molecules move faster and some move slower, but the point is that without our gravity air molecules would wander away from us and out into space. *Thus gravity is important just to keep the air around us.*
- ❑ *As gravity hugs the blanket of air to the Earth's surface, what physicists call a density gradient is set up in the air.*
- ❑ The air near the ground is pulled on by gravity and compressed by the air higher in the sky.
- ❑ *This causes the air near the ground to be denser and at a greater pressure than air at higher elevations.*



# How does gravity affect the wind?

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- ❑ Now, consider yourself at the beach (perhaps a delightful prospect!) The sun is out and the ground is getting hot. The air at the ground thus warms up and expands. As air expands the density decreases so it has a tendency to rise.
- ❑ As it rises it does not leave a void. Cooler air over the cooler ocean moves inland to replace the rising air.
- ❑ Thus a sea breeze flows inland from the ocean, driven by a density gradient caused by the combination of gravity, pressure, and temperature effects.
- ❑ At night the reverse happens. The land cools quickly relative to the ocean, and air over the ocean is warmer and rises.

# How does gravity affect the wind?

- ❑ Air moves away from the land and over the ocean to replace the rising warm ocean air and we have a land breeze flowing from the land to the sea.
- ❑ Of course, there are other factors that drive our major winds. Different areas of the Earth possess different temperatures and the winds interact with ground features.
- ❑ *The winds on our Earth are the result of a complex interplay between temperature, pressure, and density gradients, and the Earth's rotation. But at its root, the density gradient that gravity helps set up is an important factor in making our winds possible.*

# *Gravitational Force*

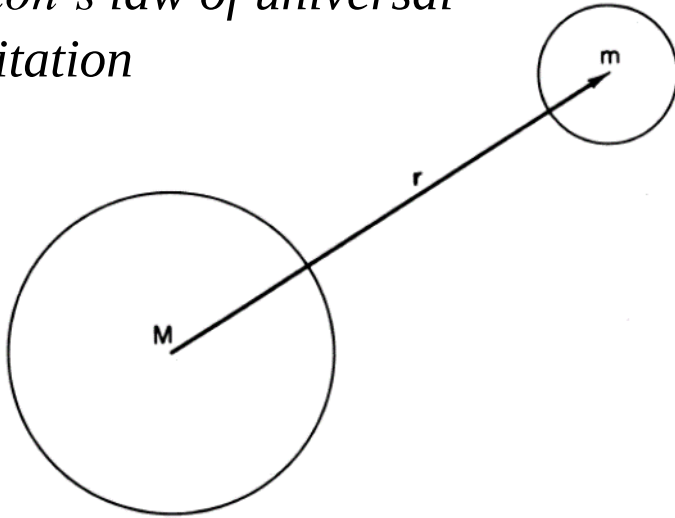
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- ❑ Newton's law of universal gravitation states that any two elements of mass in the universe attract each other with a force proportional to their masses and inversely proportional to the square of the distance separating them.
- ❑ Thus, if two mass elements  $M$  and  $m$  are separated by a distance  $r \equiv |\mathbf{r}|$  (with the vector  $\mathbf{r}$  directed toward  $m$  as shown in Fig below , then the force exerted by mass  $M$  on mass  $m$  due to gravitation is

# Gravitational Force

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*Newton's law of universal gravitation*



$F_g = -\frac{GMm}{r^2} \left(\frac{\mathbf{r}}{r}\right)$  ..... the force exerted by mass  $M$  on mass  $m$  due to gravitation

$$\frac{F_g}{m} = g^* = -\frac{GM}{r^2} \left(\frac{\mathbf{r}}{r}\right)$$

$M$  = mass of the earth

$m$  = mass element of the atmosphere

$G$  is the gravitational constant

❑ The earth is much more massive than any air parcel and the distance between the center of the earth and a parcel at the top of the troposphere versus the bottom of the troposphere is so small, that for weather applications, the magnitude of the gravitational force ( $g$ ) acting on any air parcel is considered to be **constant** throughout the troposphere:  **$9.8 \text{ m} / \text{sec}^2$** .

# Exercise

- What is the force of gravity that is exerted on the earth by the sun? the mass of the sun and the earth are  $1.99 \times 10^{30} \text{kg}$  and  $5.97 \times 10^{24} \text{kg}$  respectively. The distance between the earth and the sun is  $1.496 \times 10^{11} \text{m}$ .

# #Atmospheric Dynamics

## *Chapter Two: Fundamental Forces and Their Derivation*



### *Learning outcome*

- To Perceive the Viscous Force and its derivation
- To Know the Coriolis force
- To identify the Centrifugal and centripetal forces

# Viscous Force

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- Any real fluid is subject to internal friction (viscosity), which *causes it to resist the tendency to flow*.
- Although a complete discussion of the resulting *viscous force* would be rather complicated, the basic physical concept can be illustrated by a simple experiment.
- A layer of incompressible fluid is confined between two horizontal plates separated by a distance  $l$  as shown in Fig. below.
- The lower plate is fixed and the upper plate is placed into motion in the  $x$  direction at a speed  $u_0$ .

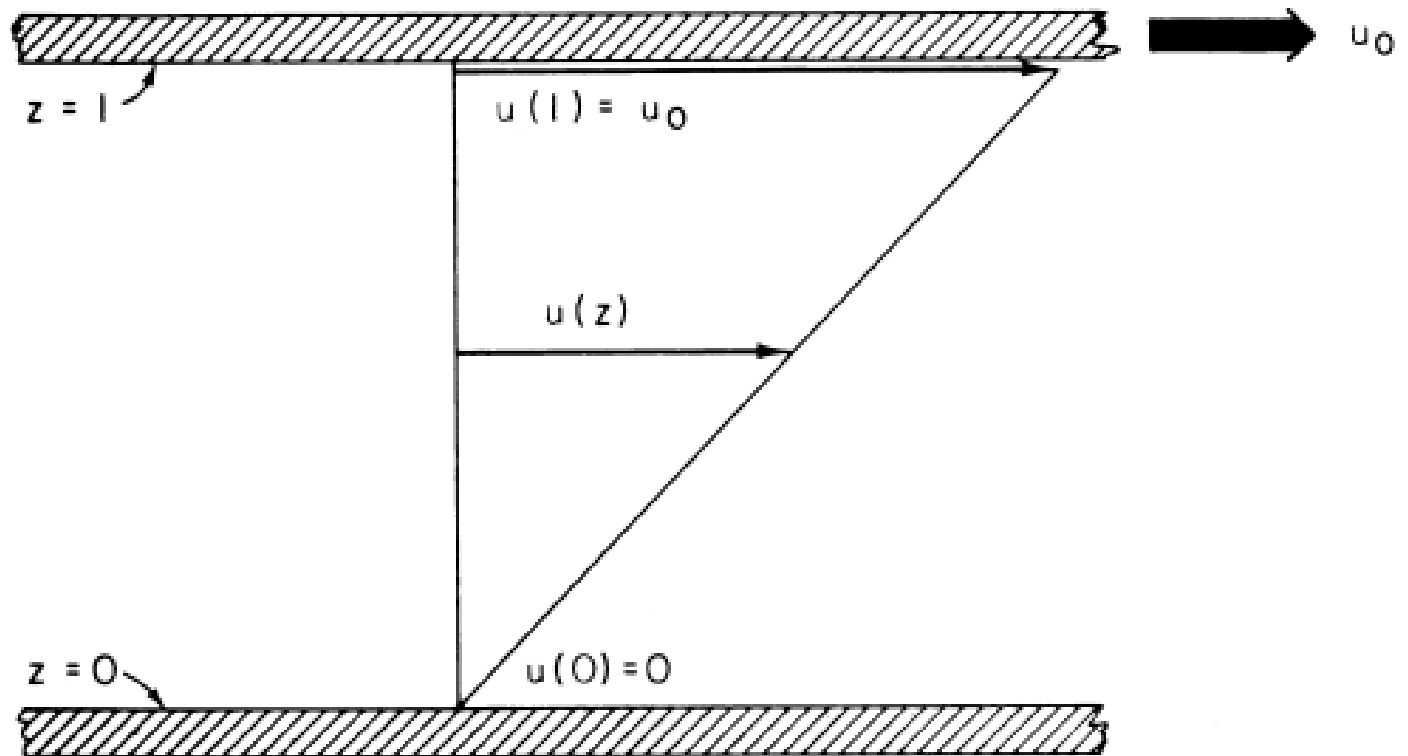


# Viscous Force

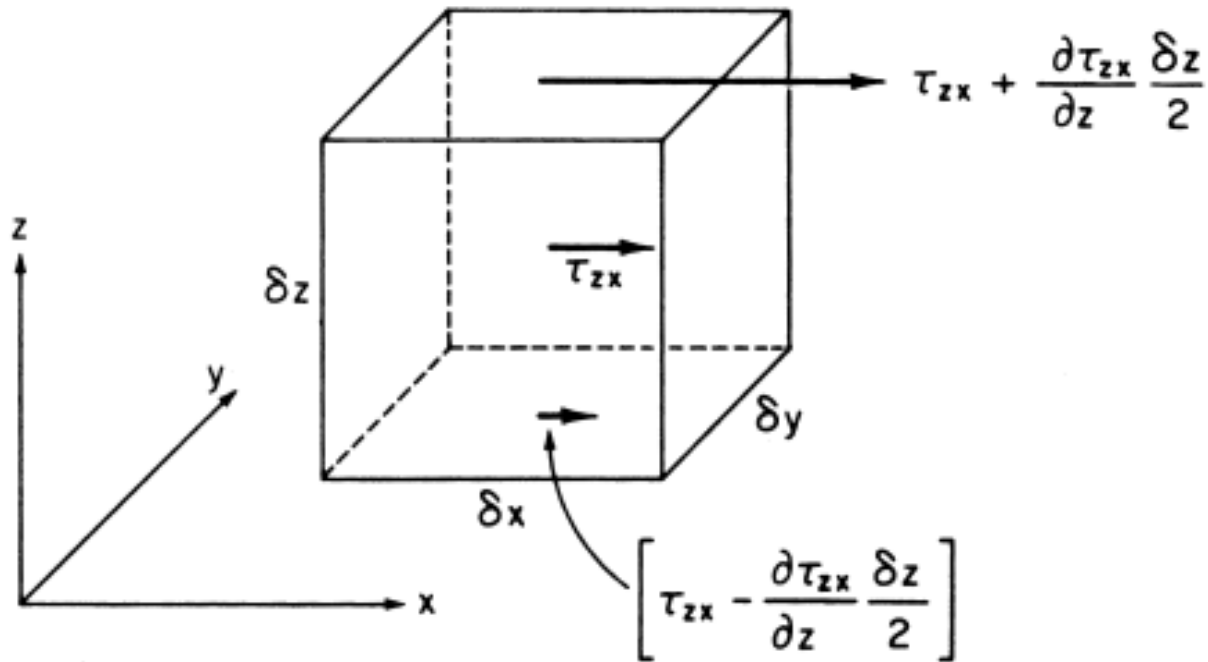
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- Viscosity forces the fluid particles in the layer in contact with the plate to move at the velocity of the plate. Thus, at  $z = l$  the fluid moves at speed  $u(l) = u_0$ , and at  $z = 0$  the fluid is motionless.
- The force tangential to the upper plate required to keep it in uniform motion turns out to be proportional to the area of the plate, the velocity, and the inverse of the distance separating the plates.
- Thus, we may write  $F = \mu A u_0 / l$  where  $\mu$  is a constant of proportionality, the *dynamic viscosity coefficient*.

❖ *The direction of the frictional force always acts in a direction opposite to the motion of the air, and therefore, always acts to reduce the speed of the flow.*



## Viscous Force derivation



Dividing this expression by the mass  $\rho \delta x \delta y \delta z$ , we find that the viscous force per unit mass due to vertical shear of the component of motion in the  $x$  direction is

## Viscous *Force derivation*

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$$\tau_{zx} + \frac{\partial \tau_{zx}}{\partial z} \frac{\delta z}{2}$$

while the stress acting across the lower boundary on the fluid *above* is

$$- \left[ \tau_{zx} - \frac{\partial \tau_{zx}}{\partial z} \frac{\delta z}{2} \right]$$

(This is just equal and opposite to the stress acting across the lower boundary on the fluid *below*.) The net viscous force on the volume element acting in the  $x$  direction is then given by the sum of the stresses acting across the upper boundary on the fluid below and across the lower boundary on the fluid above:

$$\left( \tau_{zx} + \frac{\partial \tau_{zx}}{\partial z} \frac{\delta z}{2} \right) \delta x \delta y - \left( \tau_{zx} - \frac{\partial \tau_{zx}}{\partial z} \frac{\delta z}{2} \right) \delta x \delta y$$

## Viscous Force derivation

$$\frac{1}{\rho} \frac{\partial \tau_{zx}}{\partial z} = \frac{1}{\rho} \frac{\partial}{\partial z} \left( \mu \frac{\partial u}{\partial z} \right)$$

- For constant  $\mu$ , the right-hand side just given above may be simplified to  $\nu \partial^2 u / \partial z^2$ , where  $\nu = \mu / \rho$  is the *kinematic viscosity coefficient*. For standard atmosphere conditions at sea level,  $\nu = 1.46 \times 10^{-5} \text{ m}^2 \text{ s}^{-1}$ .
- The resulting frictional force components per unit mass in the three Cartesian coordinate directions are

$$F_{rx} = \nu \left[ \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} \right]$$

$$F_{ry} = \nu \left[ \frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} + \frac{\partial^2 v}{\partial z^2} \right]$$

$$F_{rz} = \nu \left[ \frac{\partial^2 w}{\partial x^2} + \frac{\partial^2 w}{\partial y^2} + \frac{\partial^2 w}{\partial z^2} \right]$$

# *Coriolis Force*

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- ❑ An apparent force that arises due to the rotation of Earth.
- ❑ The Coriolis force acts to the *right of the wind in the Northern Hemisphere* and to the left of the wind in the Southern Hemisphere

The magnitude of the Coriolis force is:

- proportional to the wind speed
- *greater at higher latitudes and zero at the equator* Because the

Coriolis force is always perpendicular to the wind it can only change the wind direction and cannot change the wind speed.

# *Coriolis Force*

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❑ The **Coriolis "Force"** does the following:

- objects will **deviate to the right** of their direction of motion in the **Northern Hemisphere** (and to the left in the Southern Hemisphere)
- affects the direction an object will move across the Earth's surface, but has no effect on its speed
- is strongest for fast-moving objects and zero for stationary objects
- is zero at the equator and a maximum at the poles



# Coriolis Force

❖ The magnitude of the Coriolis force can be expressed as:

$$\left| \frac{F_{CF}}{m} \right| = |-fM|$$

where  $M$  is the wind speed ( $\text{m s}^{-1}$ ) and  $f$  is the Coriolis parameter given by:  $f = 2\Omega \sin \phi$  ( $\text{s}^{-1}$ )

$\Omega$  is the rotation rate of the planet ( $= 7.292 \times 10^{-5} \text{ s}^{-1}$ ) and

$\phi$  is the latitude (positive for Northern Hemisphere, negative for Southern Hemisphere). The zonal and meridional components of the Coriolis force are given by:

$$\frac{F_{x CF}}{m} = fv$$

$$\frac{F_{y CF}}{m} = -fu$$

$$\begin{aligned} \left( \frac{Du}{Dt} \right)_{Co} &= 2\Omega v \sin \phi = fv \\ \left( \frac{Dv}{Dt} \right)_{Co} &= -2\Omega u \sin \phi = -fu \end{aligned}$$

## *Centrifugal force Vs. centripetal force*

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- ❑ Consider a student standing at the center of a table that is rotating with angular velocity  $\omega$  in a laboratory (Figure below).
- ❑ He conducts an experiment in which he places an object of mass  $m$  at some distance  $r$  from him on the table. He finds that, if the forces of friction between the object and the table are negligible, the object moves radially away from him.

## Centrifugal force Vs. centripetal force

- ❑ Then, the student attaches a string of length  $r$  to the object, and measures the force required to hold the object in place.
- ❑ Using a series of measurements using different values of  $m$  and  $r$ , the student reports that the force required to hold the object in place acts radially toward to him and can be described by the expression  $m r \omega^2$ .
- ❑ Since the object is at rest from the point of view of the student, he concludes that, in order for there to be no net force, there must be an equal and opposite force acting radially outward.
- ❑ This outward radial force is called the **centrifugal force**.

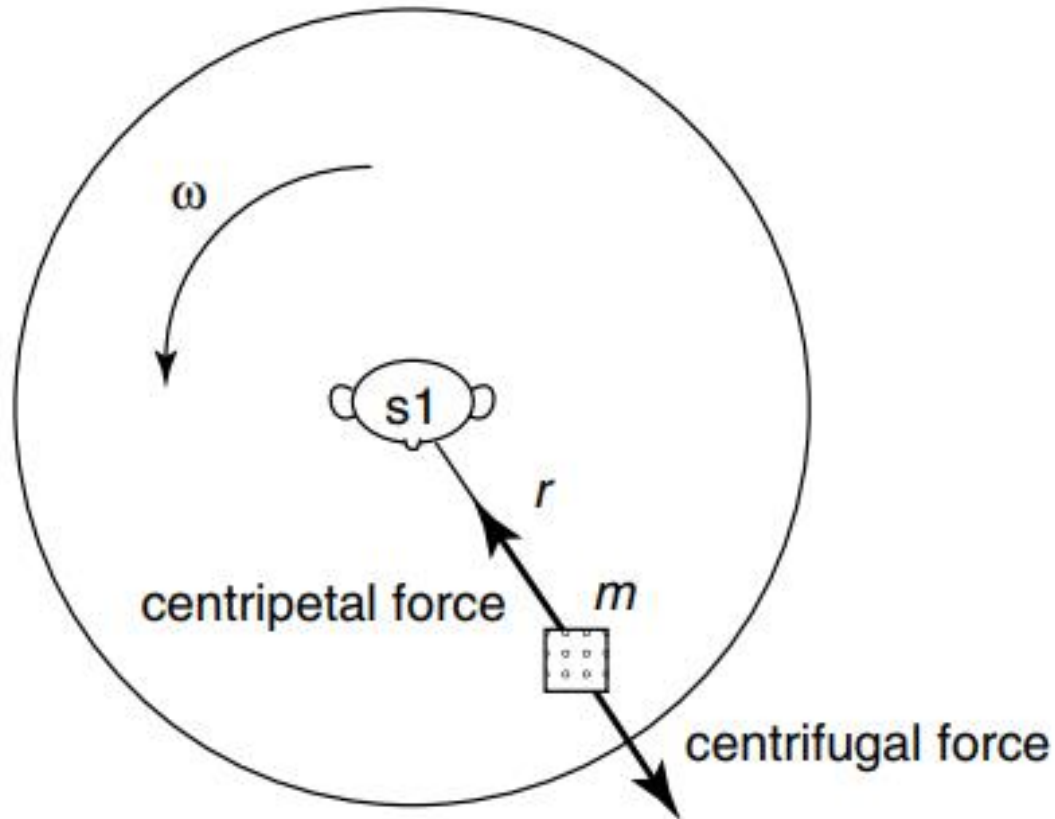
## Centrifugal force Vs. centripetal force

- ❑ Meanwhile, a second student is observing the experiment while standing on the floor beside the table.
- ❑ Her observations are quite different. She notes that the object, while being held by the string, is in fact accelerating because it is rotating at the same rate as the table.
- ❑ Hence, in her frame of reference, there is a net force acting on the object, exerted by the student on the table.
- ❑ This force acting radially inward from the object is called the **centripetal force** and can be described by the expression

$$m r \omega^2.$$

# *Centrifugal force Vs. centripetal force*

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....END

Thanks !

# Exercise

1. From the following one is **NOT** a fundamental force
  - A. Pressure gradient force
  - B. Coriolis force
  - C. The gravitational force
  - D. The friction force
2. Which one of the following is an example of the Eulerian frame of reference
  - A. Meteorological Stations
  - B. Aircraft
  - C. Meteorological balloon
  - D. All
3. What is the force of gravity that is exerted on the Earth by the sun? The mass of the sun and the earth are  $1.99 \times 10^{30} \text{kg}$  and  $5.97 \times 10^{24} \text{kg}$  respectively. The distance between the Earth and the sun is  $1.496 \times 10^{11} \text{m}$ .
  - A.  $9.52 \times 10^{22} \text{ N}$
  - B.  $7.31 \times 10^{22} \text{ N}$
  - C.  $3.52 \times 10^{22} \text{ N}$
  - D.  $6.52 \times 10^{22} \text{ N}$

4. .... Is An apparent force that arises due to the rotation of the Earth
- A. Coriolis force
  - B. Centrifugal force
  - C. The friction force
  - D. Gravitational force
5. The forces acting in the horizontal direction are the
- A. centrifugal force, the viscous force, and the Coriolis force
  - B. pressure gradient force, the viscous force, and the Coriolis force
  - C. pressure gradient force, the viscous force, and the Gravitational force
  - D. All
6. The force acting radially inward from the object is called .....
- A. Centrifugal force
  - B. Gravitational force
  - C. Coriolis force
  - D. Centripetal force
7. What are the fundamental physical principles that govern Atmospheric motions
- A. conservation of soil, conservation of momentum, and conservation of energy
  - B. conservation of mass, conservation of momentum, and conservation of energy
  - C. conservation of water, conservation of momentum, and conservation of energy
  - D. all